

Use-Case Flow Specification

PES1UG24CS125 | Charan M | Section B | Problem Statement #15 — Pharmacy Expiry & Re-order Dispatch Engine

Use Case: UC-01 — Dispense Medicine (FEFO Checkout)

Primary Actor: Pharmacy Clerk

Includes: UC-02 — Apply FEFO Picking Logic

Preconditions

1. The Pharmacy Clerk is logged into the Pharmacy Stock Management System.
2. At least one unexpired batch of the requested medicine exists in inventory.

Postconditions

1. The dispensed quantity is deducted from the selected batch's stock level.
2. A dispensing transaction record (medicine, batch number, quantity, clerk ID, timestamp) is stored in the audit trail.
3. If resulting stock falls below the safety threshold, a purchase order is automatically triggered (UC-04).

Main Success Scenario

1. Pharmacy Clerk selects “Dispense Medicine” and enters the medicine name and quantity requested.
2. System retrieves all unexpired batches of that medicine and invokes UC-02 (Apply FEFO Picking Logic) to rank them by expiry date.
3. System proposes the batch with the nearest expiry date and displays batch number, expiry date, and available quantity.
4. Pharmacy Clerk confirms the proposed batch.
5. System validates that the requested quantity does not exceed the batch's available stock.
6. System deducts the dispensed quantity from the batch and updates the inventory in real time.
7. System logs the transaction (medicine, batch number, quantity, clerk ID, timestamp) to the audit trail.
8. System checks the medicine's remaining total stock against its safety threshold.
9. System displays a dispensing confirmation and prints/emails a receipt to the clerk.
10. Use case ends successfully.

Alternate Flow

5a. Requested Quantity Exceeds Batch Stock

- a) System displays an “Insufficient batch quantity” message and shows the next-nearest-expiry batch with available stock.
- b) Pharmacy Clerk chooses to split the order across batches or reduce the requested quantity.
- c) If the clerk selects a batch other than the FEFO-proposed one, System requires a justification note (FR-004) before proceeding.
- d) Flow resumes at step 5 of the Main Success Scenario with the newly selected batch.

8a. Stock Falls Below Safety Threshold

- a) System automatically invokes UC-04 (Generate Purchase Order) for the affected medicine.
- b) System notifies the Pharmacy Clerk that a re-order has been triggered.
- c) Flow resumes at step 9 of the Main Success Scenario.